

Combined anthropogenic and climate pressures on the Kura River and their implications for water security, ecosystem sustainability and green economic resilience

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Keywords

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ABSTRACT

The Kura River is a strategic water artery for Azerbaijan, supporting drinking-water supply, irrigated agriculture, energy production and aquatic and riparian ecosystems. This study integrates two independently prepared 2026 research projects to examine how anthropogenic pressures and climate-related water stress interact and what these interactions imply for water security, ecosystem sustainability and green economic resilience. The first source project used desk research, official statistics, visual observation, teacher interviews and map-based comparison to assess human impacts on the river. The second combined desk research with an exploratory survey of 25 students and family members. Survey results showed that 92% considered the Kura important for Azerbaijan, 84% recognized ecological problems, 76% reported practicing water conservation and 96% supported stronger protection measures. The source materials consistently identified inefficient irrigation, untreated wastewater, solid waste, declining water availability, climate variability and riparian degradation as interconnected pressures. Recent official and scientific evidence supports the direction of these concerns, although one source-project estimate of a 35–40% flow decline could not be independently verified from the underlying data and is therefore not treated as a measured result. The analysis shows that water efficiency, pollution control, climate adaptation, riparian restoration and transboundary cooperation are not only environmental priorities but also elements of green economic resilience because they protect agricultural productivity, ecosystem services and the reliability of water-dependent livelihoods. Integrated river-basin management is therefore essential for aligning ecological protection with resource-efficient development.

INTRODUCTION

The Kura River is the largest river system in Azerbaijan and one of the principal water arteries of the South Caucasus. Its functions extend well beyond hydrology: the river supports municipal water supply, irrigated agriculture, hydropower, fisheries, riparian habitats and economic activity across the central lowlands. This multifunctional role means that deterioration in either water quantity or quality can propagate through food production, public services, ecosystems and local livelihoods. Azerbaijan is particularly exposed because a large share of its surface-water resources forms outside the country. The Ministry of Ecology and Natural Resources reports that 67–70% of national river-water resources originate in neighboring states, while recent World Bank assessments similarly describe strong dependence on transboundary inflows (Ministry of Ecology and Natural Resources, n.d.; World Bank, 2024).

Water stress in the Kura basin is not caused by a single driver. Climate change affects snow accumulation, runoff timing, drought frequency and evaporative demand, while human activities alter water withdrawals, pollution loads, riverbank condition and the efficiency with which water is delivered to farms and settlements. The 2024 Initial National Adaptation Plan of Azerbaijan notes that flows of the Kura and Aras have declined sharply in recent years and identifies safe

water supply, ecosystem protection and efficient use as major adaptation priorities (Republic of Azerbaijan, 2024). At the same time, official national monitoring indicates long-term declines in river flow and increasing competition for water as population, irrigation and economic activity expand (Ministry of Ecology and Natural Resources, n.d.).

These pressures have a direct economic dimension. Agriculture is a central component of Azerbaijan's non-oil economy and is highly exposed to drought and water scarcity. The World Bank's Country Climate and Development Report emphasizes that climate-proofing agriculture requires improved irrigation efficiency and climate-smart practices, linking water management to economic diversification and resilience (World Bank, 2023). In this paper, green economic resilience is understood as the capacity of water-dependent production and communities to withstand environmental shocks while improving resource efficiency and protecting natural capital. Under this framing, conserving Kura water and ecosystems is not separate from economic development; it is part of the infrastructure and natural-resource base on which sustainable growth depends.

The present study combines two original school-level research projects that approached the same problem from complementary directions. Ahgazade examined anthropogenic effects on the Kura ecosystem through desk research, official information, visual observation,

interviews with teachers and map-based comparison. Nurulu focused on the significance of Kura water resources, ecological problems and conservation attitudes and conducted a small exploratory survey. The objective is to preserve the evidence and reasoning of both projects while placing them in a more rigorous academic framework. The analysis asks three questions: Which anthropogenic and climate pressures are emphasized in the source research? What does the exploratory survey reveal about awareness and conservation attitudes? How do these pressures affect water security, ecosystem services and green economic resilience?

LITERATURE REVIEW

The Kura–Aras system is frequently described as a coupled transboundary water-security problem because river flow, water quality, agricultural withdrawals and governance interact across national borders. Suleymanov (2025) argues that stronger basin institutions, transparency and cooperation are needed to improve water-resource management in a context of scarcity and uneven trust among riparian states. The World Bank (2024) similarly identifies transboundary dependence, irrigation demand, infrastructure condition, water pollution and climate variability as mutually reinforcing constraints on Azerbaijan’s water security.

Climate change adds stress to already intensively used water systems. Azerbaijan’s Initial National Adaptation Plan reports increasing concern over reduced river flows and future water availability, particularly in arid lowland areas where water-intensive agriculture is concentrated (Republic of Azerbaijan, 2024). Research in the Kur–Araz lowland also demonstrates that climate and water management can interact through soil salinity and agricultural productivity. Han et al. (2024) found that future rainfall decline could contribute to renewed salinization, creating risks for agricultural sustainability and food security. This illustrates why river-flow management, drainage and climate adaptation need to be considered together rather than as separate environmental issues.

Anthropogenic water-quality pressure is another major concern. Aliyev and Abduev (2025) documented downstream changes in mineralization and major-ion composition in the Kura River, including the influence of the Aras confluence and more mineralized tributary waters. Such evidence supports the source projects’ broader concern that pollution and reduced flows can weaken the suitability of river water for ecosystems and economic uses. Lower flow can also reduce dilution capacity, making the ecological consequences of wastewater and agricultural drainage more difficult to manage.

From a green-economy perspective, water efficiency is a productivity and resilience issue. The World Bank (2023) notes that agriculture contributes less than 8% of Azerbaijan’s GDP but accounts for 36% of total employment, making climate and water shocks socially important even when the sector’s GDP share is comparatively modest. The same report recommends irrigation efficiency and climate-smart agriculture to protect productivity under higher temperatures and lower water availability. The National Strategy on Efficient Use of Water Resources for 2024–2040 further emphasizes integrated water management, infrastructure modernization, reuse, drought preparedness and ecosystem protection (President of the Republic of Azerbaijan, 2024).

The literature therefore supports an integrated conceptual pathway in which climate variability and anthropogenic pressure affect river flow and water quality; these changes influence habitats, irrigation reliability and ecosystem services; and the resulting environmental stress affects economic productivity and social resilience. The original student research is evaluated within this pathway, with clear

separation between source-derived observations and externally verified evidence.

MATERIALS AND METHODS

The study used an integrative qualitative design based on two independently prepared 2026 research dossiers. Ahgazade’s project examined the impact of human activity on the river ecosystem using online research and statistical review, official environmental information, visual observation of riverbank conditions, discussions with geography, biology and chemistry teachers, and comparison of maps and satellite imagery. The project identified inefficient irrigation, industrial and municipal wastewater, solid waste, declining flow and climate change as key pressures. It also reported a 35–40% reduction in flow near the Kura mouth relative to a stated norm; however, the underlying time series, station, baseline and calculation were not provided in the source file. That percentage was therefore retained as a source observation but excluded from verified quantitative findings.

Nurulu’s project combined desk research with an exploratory survey of 25 students and family members. Four questions asked whether respondents understood the Kura’s importance for Azerbaijan, believed the river faced ecological problems, practiced water conservation and supported stronger river-protection measures. The source-reported results were 92%, 84%, 76% and 96%, respectively. Because the sample was small and convenience-based, the survey is used only to describe the awareness pattern within the project group; no population-level inference is attempted.

For synthesis, both dossiers were coded into five analytical categories: (1) water abstraction and irrigation efficiency; (2) pollution and waste; (3) climate-related water availability; (4) ecosystem and riparian degradation; and (5) governance, awareness and transboundary dependence. A targeted literature review was then used to triangulate the direction of the source findings and to add an economic interpretation. Priority was given to recent official documents, peer-reviewed research and institutional assessments relevant to Azerbaijan. Green economic implications were assessed qualitatively through effects on agricultural productivity, ecosystem services, resource efficiency, water-treatment needs and the resilience of water-dependent livelihoods. No monetary valuation or cost-benefit analysis was performed because the source materials did not contain the data required for such estimates.

RESULTS AND DISCUSSION

Interacting anthropogenic and climate pressures

Both source projects portray the Kura as a system exposed to multiple pressures rather than a single isolated problem. Ahgazade places inefficient agricultural water use, wastewater discharge, household waste and weak riverbank management alongside climate-related reductions in water availability. Nurulu similarly identifies overuse, pollution, climate change and inefficient management as risks to future water security. The combined interpretation is important because the same amount of pollution can create greater ecological stress when river flow is low, while inefficient withdrawals can amplify the consequences of drought. In other words, anthropogenic and climatic pressures can reinforce one another.

The direction of the source findings is supported by official monitoring. The Ministry of Ecology and Natural Resources reports that 67–70% of Azerbaijan’s river-water resources form outside the country and that water resources of the Kura and Aras are reduced by upstream use. Its monitoring summary also indicates that, compared with 1961–1990, average flow in transboundary rivers during 1991–2023 declined by approximately 9.1–21%, with even larger

reductions recorded for some rivers over the most recent decade (Ministry of Ecology and Natural Resources, n.d.). These national indicators do not verify Ahgazade's specific 35–40% estimate at the Kura mouth, but they support the broader conclusion that declining flow is a real management concern.

Pollution constitutes the second pathway of interaction. The source projects describe untreated municipal and industrial wastewater, agricultural chemicals and litter as threats to river quality and aquatic life. Recent peer-reviewed evidence shows measurable downstream changes in Kura mineralization and ion composition, confirming that water quality varies significantly along the system (Aliyev & Abduev, 2025). From a management perspective, protecting water quality increases the usable share of a scarce resource. Pollution control can therefore reduce pressure on treatment systems and help preserve the ecological functions that support fisheries, agriculture and riverine habitats.

Table 1 translates the main pressures retained from the student research into environmental and green-economy pathways. The table does not assign causal percentages. Instead, it clarifies how actions that improve water efficiency or ecosystem condition can simultaneously support climate adaptation, natural-capital protection and resource-efficient development.

Environmental awareness and conservation attitudes

Nurulu's survey provides a small but internally consistent picture of awareness. Among 25 respondents, 92% considered the Kura important for Azerbaijan and 84% believed that the river faced ecological problems. Seventy-six percent reported that they save water, while 96% supported stronger measures to protect the river. The gap between near-universal support for stronger protection and the lower share reporting personal conservation behavior suggests that environmental concern does not always translate completely into everyday practice. Because the respondents were students and family members selected for a school research project, this pattern should not be generalized to the population of Sumgait or Azerbaijan.

The survey nevertheless adds a social dimension that is often absent from purely hydrological assessments. Water-saving technologies require user acceptance, while pollution prevention depends partly on household, agricultural and industrial behavior. Public awareness can therefore complement infrastructure investment, although it cannot substitute for effective wastewater treatment, flow measurement or irrigation modernization. Green economic resilience depends on both: efficient infrastructure reduces physical losses, and informed users reduce avoidable demand and support long-term conservation measures.

Figure 1 presents the four source-reported survey indicators. The strongest response was support for additional Kura-protection measures (96%), followed by recognition of the river's national importance (92%). Recognition of ecological problems was also high (84%), while self-reported water-saving behavior was lower at 76%. This hierarchy is useful for designing education and citizen-engagement activities because it suggests that awareness is already substantial within the small study group, whereas consistent behavior may require more practical guidance, incentives and visible institutional action.

Table 1. Integrated pressure pathways identified in the source research

Pressure pathway	Source observation and green-economy relevance
Irrigation inefficiency	Outdated irrigation practices and avoidable losses — higher water productivity and more resilient agricultural output.
Pollution and waste	Wastewater, agricultural chemicals and solid waste threaten river quality — pollution control can protect fisheries.

Pressure pathway	Source observation and green-economy relevance
	irrigation and natural capital while lowering treatment pressure.
Climate-related scarcity	Reduced flow, heat and rainfall variability increase water stress — climate-resilient allocation can lower production risk.
Riparian degradation	Loss of tugay vegetation and riverbank pressure weaken ecosystem condition — restoration protects ecosystem services, erosion control and biodiversity.
Transboundary dependence	A large share of national river flow originates outside Azerbaijan — cooperation and shared monitoring can reduce resource-security risk.

Implications for ecosystem sustainability and water security

Ecosystem sustainability is directly connected to water security because riparian vegetation, wetlands and aquatic habitats contribute to erosion control, water purification, biodiversity and the stability of riverine landscapes. Ahgazade's source project specifically emphasized restoration of tugay forests and riverbank vegetation, while Nurulu highlighted the role of the Kura ecosystem in supporting fish, birds and other species. These observations align with a nature-based interpretation of water security: protecting ecosystem condition helps maintain natural functions that engineered systems would otherwise need to replace at additional cost.

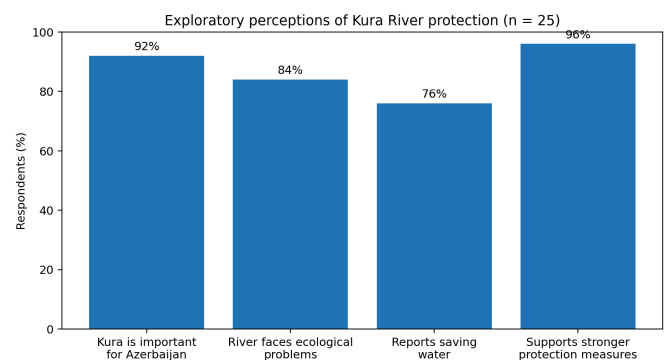


Figure 1. Source-reported environmental awareness and water-conservation responses (n = 25)

Climate pressure also influences ecosystem condition through lower flow, higher temperatures and altered seasonal water availability. Azerbaijan's adaptation planning recognizes that reduced Kura and Aras flows can make it more difficult to supply agriculture during hot months and can intensify future water-security risks (Republic of Azerbaijan, 2024). Han et al. (2024) further show that climate and water management interact with salinity in the Kur–Araz lowland. These links demonstrate that biodiversity protection, soil management, drainage and efficient irrigation should be viewed as connected components of basin resilience rather than independent sectoral tasks.

Green economic resilience and resource-efficient adaptation

The economic relevance of Kura resilience is most evident in agriculture. Water scarcity can reduce irrigation reliability, raise production risk and accelerate soil degradation, while inefficient conveyance means that part of the abstracted resource produces no economic value. The World Bank (2023) characterizes agriculture as important to Azerbaijan's non-oil economy and employment and recommends higher irrigation efficiency and climate-smart practices. For the Kura basin, this implies that drip and sprinkler systems where appropriate, canal rehabilitation, measurement of withdrawals and better scheduling can be interpreted not only as water-saving interventions but also as green-economy investments that improve the productivity of each unit of water used.

Integrated management priorities and study limitations

Five management priorities follow from the synthesis. First, irrigation modernization should reduce avoidable abstraction and increase water productivity. Second, wastewater treatment and pollution control should protect the quality of the remaining resource. Third, climate adaptation should combine drought planning, flow monitoring and flexible allocation with soil- and salinity-management measures. Fourth, riparian restoration should protect natural capital and ecosystem services. Fifth, transboundary information exchange and basin-scale cooperation should be strengthened because a large share of Azerbaijan's water originates outside national borders. These priorities are broadly consistent with the National Strategy on Efficient Use of Water Resources for 2024–2040 (President of the Republic of Azerbaijan, 2024).

The study has several limitations. The research integrates two school-level projects rather than a standardized basin-monitoring dataset. Ahgazade's visual observations and reported statistics were not accompanied by georeferenced field records or raw hydrological data, and the specific 35–40% flow-decline estimate could not be independently reconstructed. Nurulu's survey had only 25 convenience respondents and therefore cannot support population inference. The green-economic analysis is qualitative because the dossiers contain no farm-level production costs, water tariffs, treatment costs or ecosystem-service valuation. Future research should combine measured discharge and water-quality data with georeferenced field observations, larger surveys and economic indicators such as crop water productivity, avoided treatment costs and the value of ecosystem services.

CONCLUSIONS

This study integrated two original research projects to assess how human activity and climate pressure jointly affect the Kura River and why those effects matter for water security, ecosystem sustainability and green economic resilience. The source materials consistently identify inefficient irrigation, untreated wastewater, solid waste, declining water availability, climate variability and degradation of riparian ecosystems as interconnected pressures. Nurulu's exploratory survey of 25 respondents found high recognition of the Kura's importance and ecological problems and very strong support for additional protection measures, although self-reported water-saving behavior was somewhat lower. External evidence supports the direction of the students' concerns: Azerbaijan is highly dependent on transboundary surface water, recent monitoring indicates declining river flows, and climate change, salinity, pollution and irrigation inefficiency create combined risks for agriculture and ecosystems. The economic implication is that protecting the Kura is also an investment in resource efficiency and resilience. Efficient irrigation, water-quality protection, climate-smart agriculture, riparian restoration and basin cooperation can safeguard natural capital while reducing vulnerability in water-dependent production and livelihoods. The article does not claim a monetary benefit from these measures, but it demonstrates a clear pathway through which ecological protection and green economic development can reinforce one another. Future work should strengthen this pathway with measured hydrological, ecological and economic data.

AUTHOR CONTRIBUTIONS

A.A. and N.N. contributed the original research materials, problem identification, source collection and preliminary analysis. N.N. contributed the exploratory survey and its descriptive results. A.J. contributed supervision of manuscript development, methodological integration, green-economy framing, review and editing. All authors contributed to interpretation, reviewed the final manuscript and approved the submitted version.

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

FUNDING

This research received no external funding.

DATA AVAILABILITY

The source-level qualitative materials and the four aggregate survey percentages used in this synthesis are reported in the article. The survey included 25 respondents. Respondent-level records, raw hydrological measurements and georeferenced observation data were not available in the source files; therefore, no inferential statistical or hydrological dataset is provided.

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